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METHOD AND APPARATUS FOR CMS HAVING TOUCHSCREEN-BASED ZOOM FEATURES

Abstract

A method for a camera monitor system (CMS) includes displaying image feeds from a plurality of CMS cameras on a plurality of CMS displays. The image feeds depict an environment surrounding a commercial vehicle. The method also includes receiving a touchscreen command for a particular one of the CMS displays on a touchscreen interface and adjusting a zoom level of the image feed on the particular one of the CMS displays based on the touchscreen command. A camera monitor system (CMS) is also disclosed.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a camera monitor system (CMS), and more particularly to a method and apparatus for a CMS having touchscreen-based zoom features.

BACKGROUND

[0002] Vehicle camera systems for mirror replacement or for supplementing mirror views are utilized in commercial vehicles to enhance the ability of a vehicle operator to see a surrounding environment of the commercial vehicle. Camera monitor systems (CMS) utilize one or more cameras to provide an enhanced field of view to a vehicle operator. In some examples, the mirror replacement systems cover a larger field of view than a conventional mirror, or include views that are not fully obtainable via a conventional mirror.

SUMMARY

[0003] A method for a camera monitor system (CMS) according to an example embodiment of the present disclosure includes displaying image feeds from a plurality of CMS cameras on a plurality of CMS displays. The image feeds depict an environment surrounding a commercial vehicle. The method also includes receiving a touchscreen command for a particular one of the CMS displays on a touchscreen interface and adjusting a zoom level of the image feed on the particular one of the CMS displays based on the touchscreen command.

[0004] In a further embodiment of the foregoing embodiment, the method includes providing the touchscreen interface on the particular one of the CMS displays.

[0005] In a further embodiment of any of the foregoing embodiments, adjusting the zoom level includes utilizing an object detection algorithm to identify at least one predefined item in one of the image feeds. The predefined item is a vehicle item or an environmental item. Adjusting the zoom level also includes augmenting the image feed on the particular one of the CMS displays to add a visual indication of the at least one detected predefined item. The touchscreen command is a selection of a particular one of the at least one detected predefined items based on the augmenting. The adjusted zoom level enlarges the selected item on one of the CMS displays.

[0006] In a further embodiment of any of the foregoing embodiments, adjusting the zoom level includes utilizing an object detection algorithm to identify at least one predefined item in one of the image feeds. The predefined item is a vehicle item or an environmental item. Adjusting the zoom level includes, based on receiving the touchscreen command within a selection area for a particular one of the predefined items, prompting the user to confirm that the user meant to select the particular one of the at least one predefined items, and upon receiving confirmation from the user that the user meant to select the particular one of the predefined items, performing the adjusting to enlarge the particular one of the at least one predefined items.

[0007] In a further embodiment of any of the foregoing embodiments, the method includes providing the touchscreen interface on one of the plurality of CMS displays that is separate from the particular one of the CMS displays.

[0008] In a further embodiment of any of the foregoing embodiments, the touchscreen command is a reverse pinch command in which two fingers that contact the touchscreen interface spread apart from each other, and the adjusting includes increasing a zoom level of the image feed on the particular one of the CMS displays; or the touchscreen command is a pinch command in which two fingers that contact the touchscreen interface move towards each other, and the adjusting includes reducing a zoom level of the image feed on the particular one of the CMS displays.

[0009] In a further embodiment of any of the foregoing embodiments, the method includes, based on a current zoom level of the image feed on the particular one of the CMS displays differing from a default zoom level, displaying a zoom notification element to a vehicle occupant on the touchscreen interface; and reverting to the default zoom level based on receipt of a touchscreen command on the touchscreen interface in which the zoom notification element is selected.

[0010] In a further embodiment of any of the foregoing embodiments, the touchscreen command in

which the zoom notification element is selected is a double click of the zoom notification element.
[0011] In a further embodiment of any of the foregoing embodiments, the method also includes, based on receipt of a selection of an object through the touchscreen interface, panning the image feed on the particular one of the CMS displays to keep the object in a field of view of the image feed on the particular one of the CMS displays as the object moves relative to the commercial vehicle.
[0012] In a further embodiment of any of the foregoing embodiments, the adjusting includes maintaining the image feed at a default zoom level as a background image in a first display area of the particular one of the CMS displays, and superimposing a copy of the image feed at the adjusted zoom level on top of the background image in a second display area of the particular one of the CMS displays.
[0013] A camera monitor system (CMS) according to an example embodiment of the present disclosure includes a plurality of CMS cameras configured to record images of an environment surrounding a commercial vehicle, a plurality of CMS displays, and a CMS electronic control unit (ECU). The CMS ECU is configured to display image feeds from the plurality of CMS cameras on the plurality of CMS displays, receive a touchscreen command for a particular one of the CMS displays on a touchscreen interface, and adjust a zoom level of the image feed on the particular one of the CMS displays based on the touchscreen command.
[0014] In a further embodiment of the foregoing embodiment, the touchscreen interface is provided on the particular one of the CMS displays.
[0015] In a further embodiment of any of the foregoing embodiments, to adjust the zoom level, the CMS ECU is configured to utilize an object detection algorithm to identify at least one predefined item in one of the image feeds. The predefined item is a vehicle item or an environmental item. The CMS ECU is also configured to augment the image feed on the particular one of the CMS displays to add a visual indication of the at least one detected predefined item. The touchscreen command is a selection of a particular one of the at least one detected predefined items based on the augmentation. The adjusted zoom level enlarges the selected item on one of the CMS displays.
[0016] In a further embodiment of any of the foregoing embodiments, to adjust the zoom level, the CMS ECU is configured to utilize an object detection algorithm to identify at least one predefined item in one of the image feeds. The predefined item is a vehicle item or an environmental item. The CMS ECU is also configured to, based on receipt of the touchscreen command within a selection area for a particular one of the predefined items, prompt the user to confirm that the user meant to select the particular one of the at least one predefined items; and, upon receipt of confirmation from the user that the user meant to select the particular one of the predefined items, perform the adjustment to enlarge the particular one of the at least one predefined items.
[0017] In a further embodiment of any of the foregoing embodiments, the touchscreen interface is provided on one of the plurality of CMS displays that is separate from the particular one of the CMS displays.
[0018] In a further embodiment of any of the foregoing embodiments, the touchscreen command is a reverse pinch command in which two fingers that contact the touchscreen interface spread apart from each other, and the adjustment includes an increase of a zoom level of the image feed on the particular one of the CMS displays; or the touchscreen command is a pinch command in which two fingers that contact the touchscreen interface move towards each other, and the adjustment includes a reduction in a zoom level of the image feed on the particular one of the CMS displays.
[0019] In a further embodiment of any of the foregoing embodiments, the CMS ECU is configured to, based on a current zoom level of the image feed on the particular one of the CMS displays differing from a default zoom level, display a zoom notification element to a vehicle occupant on the touchscreen interface, and revert to the default zoom level based on receipt of a touchscreen command on the touchscreen interface in which the zoom notification element is selected.
[0020] In a further embodiment of any of the foregoing embodiments, the touchscreen command in

which the zoom notification element is selected is a double click of the zoom notification element.
[0021] In a further embodiment of any of the foregoing embodiments, the CMS ECU is configured to, based on receipt of a selection of an object through the touchscreen interface, pan the image feed on the particular one of the CMS displays to keep the object in a field of view of the image feed on the particular one of the CMS displays as the object moves relative to the commercial vehicle.
[0022] In a further embodiment of any of the foregoing embodiments, to adjust the zoom level of the image feed on the particular one of the plurality of CMS displays, the CMS ECU is configured to maintain the image feed at a default zoom level as a background image in a first display area of the particular one of the CMS displays and superimpose a copy of the image feed at the adjusted zoom level on top of the background image in a second display area of the particular one of the CMS displays.
[0023] The embodiments, examples, and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] The disclosure can be further understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
[0025] FIG. **1** is a schematic front view of a commercial truck with a camera mirror system (CMS) used to provide at least Class II and Class IV views.
[0026] FIG. **2** is a schematic birds-eye view of the commercial truck of FIG. **1** with a CMS providing Class II, Class IV, Class V, Class VI, and Class VIII views.
[0027] FIG. **3** is a schematic top view of an example vehicle cabin interior.
[0028] FIG. **4** is a perspective view of the vehicle cabin interior of FIG. **3**.
[0029] FIG. **5**A is a schematic view of a first electronic display from FIGS. **1-2**.
[0030] FIG. **5**B is a schematic view of a second electronic display from FIGS. **1-2**.
[0031] FIG. **6**A is a schematic view of the first electronic display from FIGS. **1-2** and **5**A that includes a zoom notification element.
[0032] FIG. **6**B is a schematic view of the second electronic display from FIGS. **1-2** and **5**B in a zoom configuration.
[0033] FIG. **6**C is a schematic view of the electronic display of FIG. **6**B in another zoom configuration.
[0034] FIG. **6**D is a schematic view of the electronic display of FIG. **6**B in another zoom configuration.
[0035] FIG. **7** is a flowchart of an example method for a CMS.
DETAILED DESCRIPTION
[0036] Schematic views of a commercial vehicle **10** are illustrated in FIGS. **1-4**. The commercial vehicle **10** includes a vehicle cab or “tractor” **12** for pulling a trailer **14**, where the trailer **14** pivots with respect to the tractor **12** during turns. Although the commercial vehicle **10** is depicted as a commercial truck with a single trailer in this disclosure, it is understood that other commercial vehicle configurations may be used (e.g., different types or quantities of trailers).
[0037] A pair of camera arms **16**A-B include a respective base that is secured to, for example, the tractor **12**. A pivoting arm is supported by the base and may articulate relative thereto. At least one rearward facing camera **20**A-B is arranged respectively on or within the camera arms **16**A-B. The

exterior cameras **20**A-B respectively provide an exterior field of view FOV.sub.EX1, FOV.sub.EX2 that each include at least one of Class II and Class IV views (FIG. **2**), which are legally prescribed views in the commercial trucking industry.

[0038] The Class II view on a given side of the commercial vehicle **10** is a subset of the class IV view of the same side of the commercial vehicle **10**. Multiple cameras also may be used in each camera arm **16**A-B to provide these views, if desired. Class II (narrow) and Class IV (wide angle) views are defined in European R46 legislation, for example, and the United States and other countries have similar drive visibility requirements for commercial trucks. Any reference to a “Class” view is not intended to be limiting, but is intended as an example of the type of view provided to a display from a particular camera.

[0039] Each camera arm **16**A-**16**B may also provide a housing that encloses electronics, e.g., a controller, that are configured to provide various features of the CMS **15**. The camera arms **16**A-B may be mounted either at a roof-mount location over the cab door (as shown), or on a door-mounted bracket or station, for example.

[0040] If video of Class V and/or Class VI views is also desired, a camera housing **16**C and camera **20**C may be arranged at or near the front of the commercial vehicle **10** to provide those views (FIG. **2**).

[0041] A backup camera **20**D may be provided which provides a field of view FOV.sub.EX3. The backup camera **20**D may be mounted at a top/centerline of the trailer, at a bumper/bed level of the trailer, or at a top-corner of the back of the trailer, for example. Alternatively, or in addition to the rear trailer camera, a “fifth wheel camera” **20**E may be provided that is mounted to a rear of the tractor **12** and that provides a field of view FOV.sub.EX4. The fifth wheel camera **20**E may be mounted anywhere between the lateral plane of the fifth wheel fixture and the top/roof edge of the tractor, for example.

[0042] FIG. **3** is a schematic top view of an example vehicle cabin interior **24**, and FIG. **4** is a perspective view of the vehicle cabin interior **24**. Referring now to FIGS. **3-4** with continued reference to FIGS. **1-2**, electronic displays **18**A-E (e.g., which may be video displays, such as LCD displays) and cameras **20**A-E are shown. The various electronic displays **18**A-E and cameras **20**A-E are part of a camera monitor system (CMS) **15**, and therefore act as CMS displays and CMS cameras. As used herein, a “CMS camera” **20** is a camera configured to record images of an environment surrounding a commercial vehicle **10**, and a “CMS display” **18** is an electronic display (e.g., an LCD) that is configured to image feeds from those cameras.

[0043] The CMS **15** includes a CMS electronic control unit (ECU) **22** that acts as a controller and includes processing circuitry that supports operation of the CMS **15**. The CMS ECU **22** is operatively connected to memory (which may include any one or combination of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, VRAM, etc.)) and/or nonvolatile memory elements (e.g., ROM, hard drive, tape, CD-ROM, etc.). The processing circuitry may include one or more microprocessors, microcontrollers, application specific integrated circuits (ASICs), or the like.

[0044] The CMS displays **18**A-B are arranged on each of the driver and passenger sides within the vehicle cab **12** on or near the A-pillars **19**A-B to display Class II and Class IV views on its respective side of the commercial vehicle **10**, which provide rear facing side views along the commercial vehicle **10** that are captured by the exterior cameras **20**A-B.

[0045] As discussed above, if video of Class V and Class VI views are also desired, the camera housing **16**C and camera **20**C may be arranged at or near the front of the commercial vehicle **10** to provide those views (FIG. **2**). In the example of FIG. **3**, additional displays **18**C-E are provided. Display **18**C is arranged in the vehicle cabin interior **24** near the top center of the windshield may be used to display the Class V and Class VI views, which are toward the front of the commercial vehicle **10**, or a backup camera view (from camera **20**D or **20**E) to the driver. Display **18**D is provided in a center console area of the vehicle cabin interior **24**, and may be used for other purposes, such as navigation, infotainment, etc. Display **18**E may be part of an instrument cluster, for example.

[0046] If video of Class VIII views is desired, camera housings can be disposed at the sides and rear of the commercial vehicle **10** to provide fields of view including some or all of the Class VIII zones of the commercial vehicle **10**. In such examples, one of the displays **18**C-E may include one or more frames displaying the Class VIII views. The displays **18**A, **18**B, **18**C face a driver region within the vehicle cabin interior **24** where an operator is seated on a driver seat.

[0047] If desired, the camera arms **16**A-B may include conventional mirrors integrated with them as well, although the CMS **15** may be used to entirely replace mirrors. In additional examples, each side can include multiple camera arms, with each arm housing one or more cameras and/or mirrors.

[0048] FIG. **5**A is a schematic view of the electronic display **18**A from FIGS. **1-4**, and FIG. **5**B is a schematic view of the electronic display **18**B from FIGS. **1-4**. Each display **18**A-B includes a respective first display area **25**A-B and a respective second display area **26**A-B. In the example of FIGS. **5**A-B, display area **25**A is configured to display a Class II view from camera **16**A, and display area **26**A is configured to display a Class IV view from camera **16**A (or an additional, wide angle camera situated on the same side of the commercial vehicle as camera **16**A). Similarly, display area **25**B is configured to display a Class II view from camera **16**B, and display area **26**B is configured to display a Class IV view from camera **16**B (or an additional, wide angle camera situated on the same side of the commercial vehicle as camera **16**B).

[0049] As will be discussed below in greater detail, the CMS **15** includes functionality for zooming in and out on CMS images based on touchscreen commands received through a touchscreen interface (e.g., pinch-based commands, such as pinch and reverse pinch, and double click commands, or predefined tap sequences). The touchscreen interface may be provided on one of the CMS displays, or through a separate touchscreen interface. In one or more embodiments a “bring your own device” configuration is supported, in which a vehicle occupant can use their own personal device (e.g., a tablet) as a CMS display and/or a touchscreen interface (e.g., a tablet).

[0050] Assume for the example below that the touchscreen interface is provided in display area **25**A of CMS display **18**A. A vehicle occupant provides a reverse pinch touchscreen command for CMS display **18**B by on display area **25**A of CMS display **18**A. The command is provided in region **50**A of the display area **25**A of CMS display **18**A, which is bounded by an area indicator **52**A (a dotted line in the example of FIG. **5**A), which corresponds to region **50**B of the display area **25**B of CMS display **18**B, which depicts a vehicle **58**, and is surrounded by an area indicator **52**B (a dotted line in the example of FIG. **5**B). Since the driver may be unable to reach the CMS display **18**B, the touchscreen interface provided by electronic display **18**A may be a convenient way to provide commands for the potentially out-of-reach CMS display **18**B.

[0051] The received command is one in which two fingers that contact the touchscreen interface spread apart from each other, which requests a zoom/enlargement of the region **50**.

[0052] FIGS. **6**A and **6**B provide schematic views of the CMS displays **18**A-B, respectively, after receipt of the enlargement command. As shown in FIG. **6**A-B, after receipt of the command, CMS display **18**B now includes an enlargement of the region **50**B in display area **26**B (such that the current zoom level differs from a default zoom level), and CMS display **18**A now includes a zoom notification element **60** to indicate that the zoom level of the region **50**B has been adjusted from a default level (e.g., a standard Class II view), to a zoomed in view in display area **26**B. In this manner, the zoom notification element **60** notifies a vehicle occupant of an adjustment from a default zoom level.

[0053] In the example above, a reverse pinch command causes the CMS **15** to increase a zoom level of the image feed on CMS display **18**B. If a pinch command is received in which two fingers contact the touchscreen interface move towards each other, the CMS **15** interprets this as a command to reduce a zoom level (e.g., to or below the default zoom level), and correspondingly performs the zoom reduction.

[0054] In the example above, the CMS display **18**A provides a touchscreen interface for CMS display **18**B, and the displays **18**A and **18**B are separate from each other. In a further example, the touchscreen interface for a particular CMS display **18** is provided on the CMS display itself. Thus, a user may provide a zoom command for CMS display **18**B on one of the display areas of CMS display **18**B, for example.
[0055] In one or more embodiments, the CMS **15** reverts to the default zoom level based on receipt of a touchscreen command on the touchscreen interface in which the zoom notification element **60** is selected (e.g., a double click) and/or through a timeout (e.g., no further zoom commands being received for a predefined time period).
[0056] FIG. **6**C is a schematic view of another example zoom enlargement configuration, in which the region **50**B is enlarged within display area **25**B in a "picture-in-picture" configuration. In this example, the CMS **15** maintains the image feed from camera **20**B at a default zoom level as a background image in a first portion **56**A of display area **25**B, and superimposes a copy of the image feed from camera **20**B at the adjusted (enlarged in the case of FIG. **6**D) zoom level on top of the background image in a second portion **56**B of display area **25**B.
[0057] FIG. **6**D is a schematic view of another example enlargement display configuration, in which the region **50**B is enlarged in the display area **25**B and fills the entire display area **25**B.
[0058] In one or more embodiments, a touch-and-drag command may be provided by a vehicle occupant to move the region **52**A and correspondingly move the region **50**B, in which a vehicle occupant touches the region **52**A and drags the region to some other area of the display area **25**A to enlarge a different portion of the displayed image. In one or more embodiments, the touch-and-drag command is provided directly on the display **18**B.
[0059] In one or more embodiments, based on receipt of a selection of an object (e.g., the vehicle **58**, a vulnerable road user, a rear tire of the trailer **14**, etc.) through the touchscreen interface (e.g., while the zoom level is greater than the default zoom level), the CMS **15** pans the image feed featuring the object to keep the object in a field of view of the image feed as the object moves relative to the commercial vehicle. This may correspond to a double click, for example. The panning may be performed in any of the image enlargements of FIGS. **6**B-D, for example. Of particular interest to an operator of the commercial vehicle **10** may be an enlarged view of one or more rear tires of the trailer **14**, so that the operator can monitor the tire(s) during turns, and avoid the trailer **14** unintentionally contacting an object (e.g., a curb).
[0060] It is understood that FIGS. **6**A-D are non-limiting examples, and that display areas having non-default zoom levels could be provided in other locations as well. Also, although zoom features are discussed in connection with a Class II view above, it is understood that they could be provided for other views, such as a Class IV and/or Class V view, and/or backup camera view.
[0061] In one or more embodiments, the CMS ECU **22** is configured to utilize an object detection algorithm to identify one or predefined vehicle items and/or predefined environmental items in one of the image feeds. The predefined vehicle items may include things of interest to a driver of a commercial vehicle, such as a rear tire location, trailer edge, etc. The environmental items may include vulnerable road users ("VRUs" such as pedestrians, cyclists, etc.), curbs, or lanes, for example. Since object detection algorithms are well-known in the art, their inner workings are not discussed in detail herein.
[0062] In one or more embodiments, the CMS ECU **22** receives a touchscreen command, performs object detection (as discussed above), based on the touch screen command being received within a selection area for a detected item (e.g., within a predefined number of pixels surrounding a boundary of the detected predefined item), the CMS ECU **22** prompts the vehicle occupant to confirm that the user meant to select the detected predefined item. Upon receiving confirmation from the user that the user meant to select the particular detected item, the adjusting is performed to enlarge the particular one of the at least one predefined items. The object detection may be performed before or after the touchscreen command is received in various embodiments.

[0063] Assume that the CMS ECU **22** has detected a rear wheel in display area **25**B, and that a vehicle occupant has provided a touchscreen command to select the rear wheel (e.g., touching the depicted rear tire on display **18**B or on a corresponding location of another touchscreen interface). The CMS ECU **22** provides a prompt to the user in display area **26**B to confirm that the user intended to select the rear wheel. However, it is understood that it could be provided elsewhere, such as along a bottom of display area **25**B.

[0064] If the user selects “Yes,” then display **18**B zooms in on the rear wheel. In one or more embodiments, the CMS ECU **22** also pans to keep the selected object in display as it moves within a field of view of the CMS **15**. In one or more embodiments, if the user selects “No” to the prompt, a prompt for another nearby item is provided, such as “Trailer edge selected?”

[0065] In one example, the prompt described above corresponds to providing an augmentation of the particular item (e.g., an overlay or boundary), and upon receiving a subsequent command to confirm the selection (e.g., another tap in the same area where the first tap was provided), the CMS ECU **22** interprets the subsequent command as confirmation.

[0066] In one or more embodiments, the CMS ECU **15** utilizes an object detection algorithm to identify a predefined item (as discussed above, such as a VRU) in one of the image feeds, and augments the image feed on one of the CMS displays to add a visual indication of the detected VRU. The augmentation may include by providing an overlay on the VRU on one of the CMS displays **18** or adding an outline to the VRU on one of the CMS displays, for example. A touchscreen command is received based on the augmented display that selects the VRU, and the CMS ECU **15** adjusts the zoom level to enlarge the VRU on one of the CMS displays **18**, and optionally also to pan the image feed to follow the VRU as the VRU moves in the image feed. The touchscreen command may be a predefined command, such as a double tap or triple tap, for example.

[0067] FIG. **7** is a flowchart of an example method **100** for a CMS **15**. The ECU **22** of the CMS **15** receives image feeds from a plurality of CMS cameras **18** that depict an environment surrounding a commercial vehicle **10** (step **102**). The CMS ECU **22** displays the various image feeds on respective ones of the plurality of CMS displays **18** (step **104**). The CMS ECU **22** receives a touchscreen command for a particular one of the CMS displays **18** on a touchscreen interface and from a vehicle occupant (e.g., a driver or passenger) (step **106**), and adjusts a zoom level of the image feed on the particular one of the CMS displays based on the command (step **108**).

[0068] As discussed above, the touchscreen command may be a pinch command in which two fingers that contact the touchscreen interface move towards each other (corresponding to a request for decreasing a zoom level) or a reverse-pinch command in which two fingers that contact the touchscreen interface spread apart from each other (corresponding to a request for increasing a zoom level).

[0069] Although the examples above focus on providing zooming on CMS display **18**B, it is understood that this is a non-limiting example, and that zooming could be provided on any CMS display using the techniques discussed herein.

[0070] Although example embodiments have been disclosed, a worker of ordinary skill in this art would recognize that certain modifications would come within the scope of the claims. For that reason, the following claims should be studied to determine their true scope and content.

Claims

1. A method for a camera monitor system (CMS), comprising: displaying image feeds from a plurality of CMS cameras on a plurality of CMS displays, wherein the image feeds depict an environment surrounding a commercial vehicle; receiving a touchscreen command for a particular one of the CMS displays on a touchscreen interface; and adjusting a zoom level of the image feed on the particular one of the CMS displays based on the touchscreen command.

2. The method of claim 1, comprising providing the touchscreen interface on the particular one of the CMS displays.
3. The method of claim 1, wherein said adjusting a zoom level comprises: utilizing an object detection algorithm to identify at least one predefined item in one of the image feeds, wherein the predefined item is a vehicle item or an environmental item; and augmenting the image feed on the particular one of the CMS displays to add a visual indication of the at least one detected predefined item; wherein the touchscreen command is a selection of a particular one of the at least one detected predefined items based on said augmenting; and wherein the adjusted zoom level enlarges the selected item on one of the CMS displays.
4. The method of claim 1, wherein said adjusting a zoom level comprises: utilizing an object detection algorithm to identify at least one predefined item in one of the image feeds, wherein the predefined item is a vehicle item or an environmental item; and based on receiving the touchscreen command within a selection area for a particular one of the predefined items: prompting the user to confirm that the user meant to select the particular one of the at least one predefined items; upon receiving confirmation from the user that the user meant to select the particular one of the predefined items, performing said adjusting to enlarge the particular one of the at least one predefined items.
5. The method of claim 1, comprising providing the touchscreen interface on one of the plurality of CMS displays that is separate from the particular one of the CMS displays.
6. The method of claim 1, wherein: the touchscreen command is a reverse pinch command in which two fingers that contact the touchscreen interface spread apart from each other, and said adjusting comprises increasing a zoom level of the image feed on the particular one of the CMS displays; or the touchscreen command is a pinch command in which two fingers that contact the touchscreen interface move towards each other, and said adjusting comprises reducing a zoom level of the image feed on the particular one of the CMS displays.
7. The method of claim 1, comprising: based on a current zoom level of the image feed on the particular one of the CMS displays differing from a default zoom level, displaying a zoom notification element to a vehicle occupant on the touchscreen interface; and reverting to the default zoom level based on receipt of a touchscreen command on the touchscreen interface in which the zoom notification element is selected.
8. The method of claim 7, wherein the touchscreen command in which the zoom notification element is selected is a double click of the zoom notification element.
9. The method of claim 1, comprising: based on receipt of a selection of an object through the touchscreen interface, panning the image feed on the particular one of the CMS displays to keep the object in a field of view of the image feed on the particular one of the CMS displays as the object moves relative to the commercial vehicle.
10. The method of claim 1, wherein said adjusting comprises: maintaining the image feed at a default zoom level as a background image in a first display area of the particular one of the CMS displays; and superimposing a copy of the image feed at the adjusted zoom level on top of the background image in a second display area of the particular one of the CMS displays.
11. A camera monitor system (CMS), comprising: a plurality of CMS cameras configured to record images of an environment surrounding a commercial vehicle; a plurality of CMS displays; and a CMS electronic control unit (ECU) configured to: display image feeds from the plurality of CMS cameras on the plurality of CMS displays; receive a touchscreen command for a particular one of the CMS displays on a touchscreen interface; and adjust a zoom level of the image feed on the particular one of the CMS displays based on the touchscreen command.
12. The CMS of claim 11, wherein the touchscreen interface is provided on the particular one of the CMS displays.
13. The CMS of claim 11, wherein to adjust the zoom level, the CMS ECU is configured to: utilize an object detection algorithm to identify at least one predefined item in one of the image feeds,

wherein the predefined item is a vehicle item or an environmental item; and augment the image feed on the particular one of the CMS displays to add a visual indication of the at least one detected predefined item; wherein the touchscreen command is a selection of a particular one of the at least one detected predefined items based on said augmentation; and wherein the adjusted zoom level enlarges the selected item on one of the CMS displays.

14. The CMS of claim 11, wherein adjust the zoom level, the CMS ECU is configured to: utilize an object detection algorithm to identify at least one predefined item in one of the image feeds, wherein the predefined item is a vehicle item or an environmental item; and based on receipt of the touchscreen command within a selection area for a particular one of the predefined items: prompt the user to confirm that the user meant to select the particular one of the at least one predefined items; upon receipt of confirmation from the user that the user meant to select the particular one of the predefined items, perform the adjustment to enlarge the particular one of the at least one predefined items.

15. The CMS of claim 11, wherein the touchscreen interface is provided on one of the plurality of CMS displays that is separate from the particular one of the CMS displays.

16. The CMS of claim 11, wherein: the touchscreen command is a reverse pinch command in which two fingers that contact the touchscreen interface spread apart from each other, and the adjustment comprises an increase of a zoom level of the image feed on the particular one of the CMS displays; or the touchscreen command is a pinch command in which two fingers that contact the touchscreen interface move towards each other, and the adjustment comprises a reduction in a zoom level of the image feed on the particular one of the CMS displays.

17. The CMS of claim 11, wherein the CMS ECU is configured to: based on a current zoom level of the image feed on the particular one of the CMS displays differing from a default zoom level, display a zoom notification element to a vehicle occupant on the touchscreen interface; and revert to the default zoom level based on receipt of a touchscreen command on the touchscreen interface in which the zoom notification element is selected.

18. The CMS of claim 17, wherein the touchscreen command in which the zoom notification element is selected is a double click of the zoom notification element.

19. The CMS of claim 11, wherein the CMS ECU is configured to: based on receipt of a selection of an object through the touchscreen interface, pan the image feed on the particular one of the CMS displays to keep the object in a field of view of the image feed on the particular one of the CMS displays as the object moves relative to the commercial vehicle.

20. The CMS of claim 11, wherein to adjust the zoom level of the image feed on the particular one of the plurality of CMS displays, the CMS ECU is configured to: maintain the image feed at a default zoom level as a background image in a first display area of the particular one of the CMS displays; and superimpose a copy of the image feed at the adjusted zoom level on top of the background image in a second display area of the particular one of the CMS displays.